Medication Safety Signal Monitor: how the data becomes a flag you can trust

Built for the hospital pharmacy safety officer's three morning questions: (1) adverse events the label does not mention, (2) active recalls, (3) what changed. Trust rule: every flag traces to its source: the label versions searched (DailyMed) and the FDA recall record open in one click; the FAERS case reports are reachable by query, not yet from her screen.

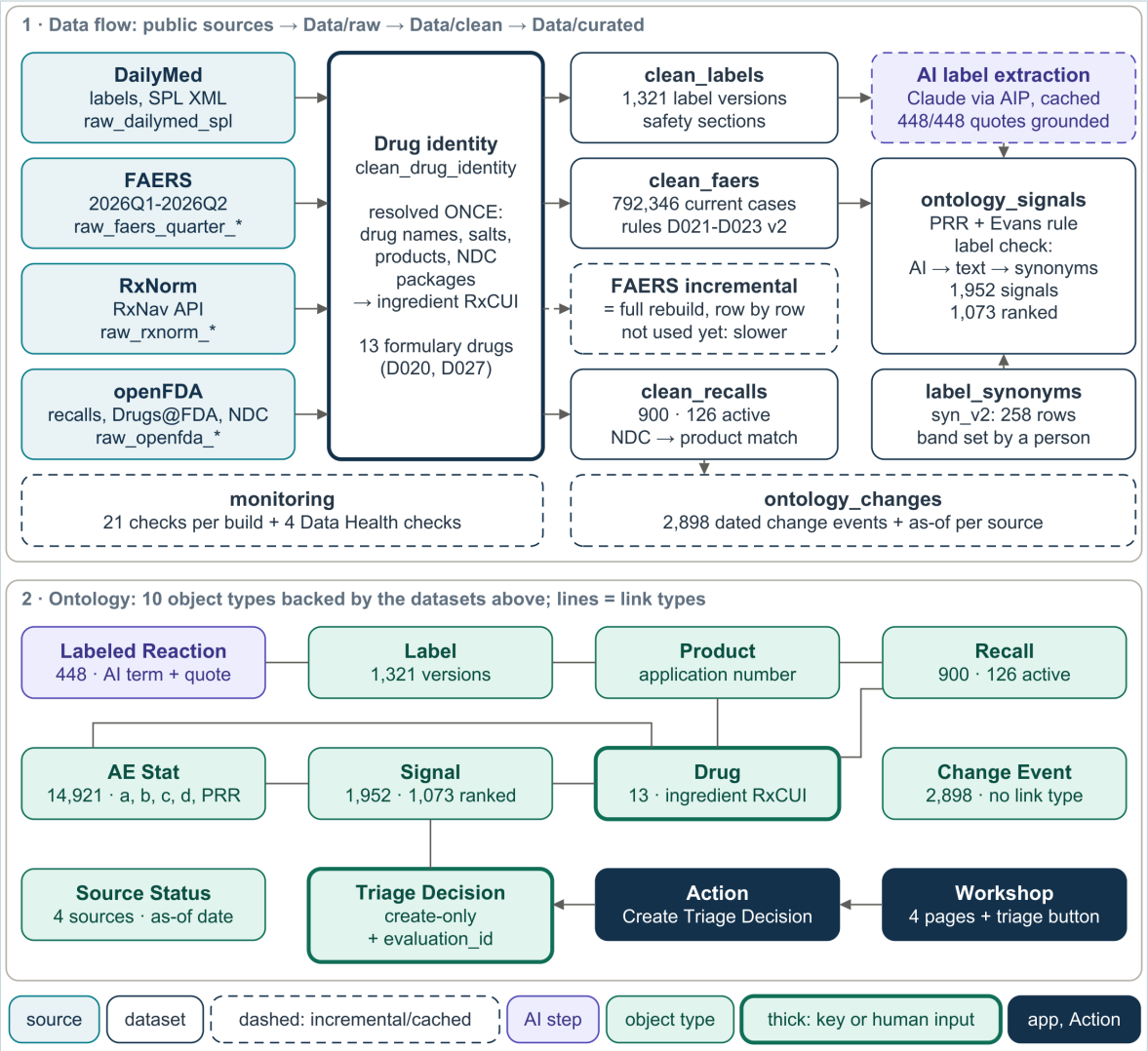


Figure. Four public sources land in Data/raw as received. Every drug name passes through one identity dataset, then clean, curated and the Ontology (Foundry's layer of objects and links the app reads). **Glossary:** FAERS = FDA adverse event reports · DailyMed = official labels (SPL = the label XML) · RxCUI = RxNorm drug ID · NDC = package code · PRR = proportional reporting ratio · Workshop = Foundry's app builder · Action = a form that writes to the Ontology.

Why these four: each decides whether a flag can be trusted, and each answers one judging row: Architecture, Data judgment, AI reliability, Application. Runners-up: the full FAERS rebuild kept as the signal input (D044, O2) and the focus buckets (D016 v3).

Four decisions that mattered most

1 Drug identity is resolved once, in the clean layer

Chose (D020; D027 proposed)**:** one dataset, `clean_drug_identity`, maps every drug name, salt, product and NDC package to an exact ingredient RxCUI. Every later step joins it.
Why: counts belong to ingredients, labels and recalls to products.
Evidence: substring matching ("morphine" also hit apomorphine and diamorphine) added 50% false morphine rows (DQ34) **OBSERVED**

2 FAERS counts each current case once, suspects only

Chose (D021, D022, D023 v2)**:** the latest case version wins and FDA deletes apply to all history; suspect drugs only, once per report; duplicate patients folded; reports listing 30+ drugs leave the main count.
Why: FAERS re-files and deletes cases; mass re-filings fake signals.
Evidence: appending quarters gave +8.7% phantom rows (DQ36). Q2 replaced 26,681 cases and deleted 655 **OBSERVED**

3 "Not in label" is a 3-step check, and the AI must quote

Chose (D037 proposed; D041, D043)**:** a FAERS term is in the label if (1) an AI term with a code-found quote matches it, else (2) its exact text is in a safety section, else (3) a person-banded synonym.
Why: each "not in label" names the label versions searched; each AI match quotes its passage.
Evidence: 448/448 quotes grounded; F1 0.97 vs proposed D015 target 0.85 (35-item Claude key, 2 labels, pharmacist pending); synonyms + wider buckets: ranked 1,355 at step 9 → 1,073 **OBSERVED**

4 Triage is its own write-back object; changes are dated

Chose (D040, D042; D045 proposed)**:** a decision is its own object, made only by an Action, tied to the signal and its `evaluation_id` (a hash of the evidence). A change feed dates each signal, recall and label change.
Why: a status column on Signal would be wiped by the next rebuild; sources run on different clocks.
Evidence: a decision survived a full rebuild (step 6 gate); 2,898 change events, 0 false ones after a rebuild **OBSERVED**

What we chose not to do

- **No MedDRA licence** (the paid dictionary of reaction terms): keyword buckets (D016 v3) and a reviewed synonym table stand in.
- **No causality, no patient data.** A signal = more reports than expected (Evans; D006 proposed) and not in the label, or only possibly (108). 1,073 ranked = a review queue, not findings.
- **Signals do not read incremental FAERS yet:** equal row for row, but 88 s vs 32 s at 2 quarters (O2). Re-measure at 4+.
- **AI kept to 2 pilot labels, and no AI judge per signal,** until a pharmacist checks the answer key.
- **No ClinicalTrials.gov, OnSIDES, daily FAERS scraping or label-change backtest** (D031 cut list; D029 proposed).
- **No build schedule** (we refresh when a build runs), **per-user bookmark or live drift** yet (drift is a replay). Known flaw: an unused duplicate Signal → Drug link.

Build facts. 13 drugs (8 opioids + 5 GLP-1s) · 4 public sources (FAERS 2026Q1-2026Q2, DailyMed, openFDA, RxNorm) · 792,346 current FAERS cases · 1,321 label versions (1,231 carry an effective date) · 900 recalls (126 active) · 1,952 signals, 1,073 ranked · 10 object types, 1 Action · Workshop v0.9.0, 4 pages · 21 monitoring checks per build. Numbers: docs/11-handin/numbers.json (captured 2026-11-05T07:50Z). Detail: diagrams 30-38, reports R01–R09, docs/06-decisions/DECISIONS.md.